

M.Sc. Robin Steiger

Date of birth: 02.06.1995

Citizenship: German

Location: Freiburg, Germany



EXPERIENCE

- **FERCHAU GmbH (SICK AG)** Waldkirch, Germany
Software Engineer Robotics Sep. 2025 - Present
 - **Manual Test Setup and Validation:** Responsible for the manual test setup and execution of safety-relevant real-world validation scenarios for mobile robots, including UR and AGV robot control and accurate pose tracking. Technologies: *ROS 2, C++, Python, UR robots, LidarLoc, RTDE*
- **Institute for Systems and Robotics** IST Lisbon, Portugal
Software Development & Research (Internship) Oct. 2023 - Oct. 2024
 - **Autonomous Robot Navigation:** Developed and tested ROS-based modules for SLAM, local planning, and improved pose estimation in mobile robotics. Technologies: *ROS, C++, Python, C*
- **JobRad GmbH** Freiburg, Germany
Software Engineer Mar. 2023 - Nov. 2023
 - **ERP System Development and Maintenance:** Developed and maintained *Odoo ERP* workflows and process automation; provided second-level support for contract management. Technologies: *Python, Odoo, SQL*
- **JW Froehlich Maschinenfabrik GmbH** Leinfelden-Echterdingen, Germany
Software Engineer & UI Design (Internship then Job) Sep. 2020 - Sep. 2021
 - **Requirements Analysis and Solution Design:** Conducted requirements analysis with cross-functional stakeholders and designed a user-friendly Windows application for company-wide document archiving and sorting. Technologies: *UML, Draw.io*
 - **WPF Development, Acceptance, and Rollout:** Developed a .NET-based WPF application from scratch; supported user acceptance, integration, and company-wide rollout. The application remains in daily use across the mid-sized company. Technologies: *VB.NET, WPF, .NET, SQL, Python, JavaScript, HTML*
- **University of Freiburg** Freiburg, Germany
Project Manager 2018, and 2021 - 2022
 - **Project Coordination and Student Mentoring:** Coordinated project work and mentored approximately 30 Computer Science students, ensuring successful project completion and harmonious team collaboration.

FOCUS AREAS

- **Autonomous Robotics:** Navigation, SLAM, localization, robot control, and real-world validation
- **Safety-Relevant Testing:** Test setup, execution, and validation quality for safety-relevant systems
- **Software Engineering:** End-to-end application development, requirements engineering, and system integration
- **Collaboration & Mentoring:** Cross-functional teamwork, stakeholder communication, and mentoring

SKILLS

- **Programming Languages:** C/C++, Python, C#, SQL, JavaScript, VB.NET
- **Robotics & Frameworks:** ROS, ROS 2, NumPy, .NET, WPF
- **Testing & Quality Assurance:** Automatic test setup, safety-relevant validation, ISTQB Foundation Level training
- **Methods:** Agile (Scrum), Requirements Engineering, UML, Test-Driven Development
- **Tools:** Git, Jira, Jama, Docker, CI/CD, Visual Studio, Linux
- **Languages:** German (C2), English and Portuguese (C1), Spanish (B2)

EDUCATION

- **University of Freiburg** Freiburg, Germany
Master of Science in Computer Science (Specialization in AI) Oct. 2021 - Feb. 2025
 - **Master Thesis:** *Adaptive Local Planning for Improved Pose Certainty in Active SLAM*
Developed and evaluated a ROS-based local planner that balances goal-directed navigation with pose correction for more reliable SLAM.
Manuscript in preparation for submission to a peer-reviewed robotics venue.
 - **Courses Completed:** Machine Learning, AI Planning, Mobile Robotics, ML in Life Science, Knowledge Representation and Reasoning, Robot Mapping, Statistical Pattern Recognition, Advanced Topics in Requirements Engineering
- **University of Freiburg** Freiburg, Germany
Bachelor of Science in Computer Science Oct. 2015 - Jun. 2021
 - **Bachelor Thesis:** *Enterprise Content Management Filing Tool*
Developed an ECM filing tool with SQL and Microsoft Outlook integration, including UX design and requirements engineering.
 - **Courses Completed:** Logic, Stochastic, Graph Theory & Optimization, Mathematics, Discrete Algebra, Programming (Python), Programming (C++), Algorithms and Data Structures, Theoretical Computer Science, Operating Systems, Computer Networks, Technical Computer Science, System Design Project, Databases and Information Systems, Software Engineering, Brain-Computer Interfaces, Information Retrieval, Artificial Intelligence, Computer Architecture, Hardware Lab, Software Lab

INTERNATIONAL & SOCIAL ENGAGEMENT

- **reha assist** Unterschneidheim, Germany
Assistance and care for a paraplegic patient 2020 - 2021
- **SWFr Freiburg** Freiburg, Germany
Support and coordination for Erasmus students 2018 - 2020
- **University of Freiburg** Freiburg, Germany
Support for a Computer Science student with Autism Spectrum Disorder 2017 - 2018
- **Sonflora** León, Nicaragua
Tutoring, organization, and care 2015
- **Nepal Schulprojekt** Dadhikot, Nepal
High School Teacher: English & Computer Science 2014